1 Solution — GIAM 6.3 Problem 7

1.1 Problem

Reflect back on the proof in Problem 5. The hypothesis "the second coordinate is nonzero" was carefully maintained throughout — that is the entire reason for restricting to $\mathbb{Z}^* = \mathbb{Z} \setminus \{0\}$. At what point in the argument did the hypothesis actually get used?

1.2 The Answer in One Line

In the **transitivity** step — specifically, in the closing line where we *cancelled* d to conclude $af = be$.

1.3 The Critical Line

Recall the transitivity argument. Given $(a, b)\, Q\, (c, d)$ and $(c, d)\, Q\, (e, f)$, so $ad = bc$ and $cf = de$, we multiplied the two equations through by f and b, chained, and reached

$$(af)\, d \;=\; (be)\, d.$$

To conclude $af = be$ — and hence $(a, b)\, Q\, (e, f)$ — we cancelled the common factor d:

$$(af)\, d \;=\; (be)\, d \;\Longrightarrow\; af \;=\; be \qquad \text{(cancellation in } \mathbb{Z}\text{)}.$$

This single step is *the* moment at which $d \neq 0$ is needed. Cancellation in $\mathbb{Z}$ — "$xz = yz$ and $z \neq 0$ together imply $x = y$" — fails outright when $z = 0$, because then $x \cdot 0 = y \cdot 0 = 0$ for every x, y, and the equation imposes no constraint on x and y.

Nowhere else in the proof is the hypothesis used. Reflexivity ($ab = ba$) and symmetry ($ad = bc \Leftrightarrow cb = da$) are pure commutativity manipulations; they hold for any ordered pair in $\mathbb{Z} \times \mathbb{Z}$, with no constraint on either coordinate. Only the cancellation step in transitivity bites.

1.4 Why the Hypothesis Has to Apply to *Every* Pair

The cancellation only needed d — the second coordinate of the *middle* pair in the chain $(a, b)\,Q\,(c, d)\,Q\,(e, f)$ — to be non-zero. So strictly speaking, the proof only requires the second coordinate of whichever pair happens to be the middle.

But Q is a single relation defined uniformly on the whole set, and *any* pair (c, d) might play the middle role in some transitivity chain. For transitivity to hold *unconditionally*, *every* pair's second coordinate must be non-zero. That is the cleanest way to package the constraint: restrict the underlying set to $\mathbb{Z} \times \mathbb{Z}^*$ so the cancellation is always available, no matter where a pair sits in a chain.

1.5 What Goes Wrong Without It

Drop the hypothesis — allow the second coordinate to be zero — and transitivity breaks. Concretely (re-using the example from Problem 5):

$$(1, 1)\,Q\,(0, 0) \quad \textbf{since} \quad 1 \cdot 0 = 1 \cdot 0 = 0,$$

$$(0, 0)\,Q\,(1, 2) \quad \textbf{since} \quad 0 \cdot 2 = 0 \cdot 1 = 0,$$

both vacuously true. But

$$(1, 1)\,Q\,(1, 2) \iff 1 \cdot 2 = 1 \cdot 1 \iff 2 = 1,$$

which is false. The chain $(1, 1)\,Q\,(0, 0)\,Q\,(1, 2)$ would force a relation that simply does not hold — and the failure is precisely the spot where we would have to cancel 0.

1.6 Remark — Cancellation Is the Whole Game

The hypothesis is really about *cancellation*, not about $\mathbb{Z}$ specifically. In any **integral domain** R (a commutative ring with no zero divisors), the same cancellation rule holds: $xz = yz$ with $z \neq 0$ implies $x = y$. Replacing $\mathbb{Z}$ by R and $\mathbb{Z}^*$ by $R \setminus \{0\}$ throughout, the entire proof of Problem 5 goes through unchanged — and yields the field of fractions $\mathbf{Frac}(R)$ as the quotient.

So integral domains are *exactly* the rings for which the field-of-fractions construction works, and the reason traces back to this single line: cancellation of the middle-coordinate d in the transitivity step.